# Reliable Evaluation of QLoRA Data Selection: A Controlled Protocol for Auditing Subset Gains

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***Abstract*—QLoRA makes repeated fine-tuning experiments practical, but small-subset gains can be easily overinterpreted as selector improvements. This paper presents a controlled evaluation protocol for auditing QLoRA data-selection claims through four checks: full-data recipe sanity, strong random and stratified-random baselines, difficulty-proxy diagnostics, and cost-reporting caveats. We instantiate the protocol on AG News, BoolQ, and GSM8K with Qwen2.5 models under QLoRA, yielding a 248-row experiment-accounting matrix. The results show that apparent subset gains change interpretation under these checks: AG News gains are recipe-sensitive, BoolQ favors full-data training, and GSM8K shows a moderate difficulty-diversity band that is close to full but interval-limited. Difficulty diagnostics further reveal label-distribution confounding on AG News and BoolQ and length confounding on GSM8K. These findings suggest that QLoRA subset gains should be reported as controlled claims supported by recipe, baseline, proxy, and cost-accounting caveats, rather than as raw selector rankings.**

***Index Terms*—QLoRA, data selection, parameter-efficient fine-tuning, evaluation protocol, subset selection, difficulty proxy, baselines**

## I. Introduction

LoRA and QLoRA make repeated adaptation experiments practical through low-rank adapters and a frozen 4-bit backbone [1], [2]. Yet a subset that outperforms one full-data run may reflect an under-tuned recipe, a strong unreported random baseline, a biased proxy, or incomplete cost accounting rather than a superior selector. Adapter settings, epochs, and token exposure make these checks especially important under QLoRA.

Selected or curated instruction data can be competitive [3]–[7], but selection quality depends on the setting and random selection can be difficult to beat [8]–[10]. The central question is therefore what evidence supports interpreting a subset-over-full observation as a selection gain.

This paper frames QLoRA data selection as an evaluation-reliability problem. It studies four research questions.

- **RQ1:** Do apparent subset-over-full gains survive full-data and subset-side recipe sanity checks?
- **RQ2:** Are random and stratified-random subsets weak baselines, or mandatory controls?
- **RQ3:** Does normalized-negative-log-likelihood (normalized-NLL) difficulty behave like an unconfounded utility proxy?
- **RQ4:** What reporting checks reduce misattribution of QLoRA subset gains?

We instantiate a controlled evaluation protocol on AG News, BoolQ, and GSM8K with Qwen2.5 [11]. It combines recipe sanity, strong simple baselines, proxy diagnostics, and cost caveats. The goal is not to certify a universal selector, but to classify observations as robust, recipe-sensitive, interval-limited, proxy-confounded, or cost-limited.

## II. Related Work

LoRA and QLoRA enable adapter-only fine-tuning through low-rank updates and a frozen 4-bit backbone [1], [2]. Data selection spans quality filtering, diversity, influence, difficulty, and task-specific scoring [8], [9]; curated subsets and methods such as LIMA, AlpaGasus, LESS, Superfiltering, D3, and Data Whisperer show that small subsets can be competitive [3]–[6], [12], [13]. Difficulty-related work also shows that high-loss or frequently forgotten examples may be noisy or atypical [14]–[17]. Reliable comparison therefore requires appropriate paired tests, strong baselines, and reporting of development and compute choices [18]–[20]. Unlike selector-ranking studies, this work audits whether an apparent subset gain survives those checks.

## III. Controlled Evaluation Protocol

### A. Problem Setup

For each task, a selection pool, development set, and evaluation split are fixed. In this paper, "full-data" means training on the entire fixed experimental selection pool, not necessarily the original full training split of the benchmark. We therefore use full-data, full-pool, and 100% budget interchangeably after this definition. A selector chooses examples from the fixed selection pool at budget $b \in \{10\%, 20\%, 40\%\}$ unless otherwise stated, while the full-data reference uses 100% of that pool. Models are fine-tuned with QLoRA and evaluated on the held-out evaluation split with task-appropriate metrics. The three tasks cover topic classification, yes/no reading comprehension, and mathematical reasoning: AG News [21], BoolQ [22], and GSM8K [23]. These tasks are not intended to exhaustively represent all QLoRA use cases; they are chosen to stress different audit risks: label-balance effects in topic classification, answer-prior effects in yes/no question answering, and length-related effects in mathematical reasoning. The development split is used for training-time validation and sanity-check workflows, while the final reported metrics are computed on the held-out evaluation split shown in Table I.

TABLE I
TASK AND EVALUATION SETUP. POOL DENOTES THE FIXED EXPERIMENTAL SELECTION POOL, NOT NECESSARILY THE ORIGINAL FULL TRAINING SPLIT OF THE BENCHMARK.

| Task | Pool | Dev | Eval | Metric | Output |
|---|---|---|---|---|---|
| AG News | 18,000 | 2,000 | 7,600 | Acc. | 4-way topic label |
| BoolQ | 8,484 | 943 | 3,270 | Acc. | yes/no answer |
| GSM8K | 6,725 | 748 | 1,319 | EM | numeric final answer |

TABLE II
MAIN QLORA AND TRAINING CONFIGURATION.

| Component | Configuration |
|---|---|
| Models | Qwen2.5-0.5B pilot; Qwen2.5-1.5B-Instruct main matrix |
| Quantization | 4-bit NF4 with double quantization |
| LoRA | rank 16, alpha 32, dropout 0.05 |
| Target modules | q, k, v, o, up, down, and gate projections |
| Optimizer | paged AdamW 8-bit |
| Fixed recipe | lr $2 \times 10^{-4}$, 3 epochs unless varied |
| Batching | per-device batch 4, gradient accumulation 8 |
| Hardware | A100 40GB |
| Seeds | data seed 42; three training seeds for main comparisons |

The analyzed matrix contains 248 completed fine-tuning/evaluation records: 26 Qwen2.5-0.5B pilot rows, 90 Qwen2.5-1.5B fixed-recipe rows, 108 additional baseline and band-ablation rows, and 24 AG News recipe-sanity rows. Each record fixes a task, model, selection condition, budget, recipe, and training seed; the 248 records are an accounting matrix rather than independent statistical samples.

In this paper, a *recipe* refers primarily to fine-tuning settings varied in the sanity grid, especially epoch count and learning rate. Unless explicitly stated, the remaining QLoRA and evaluation settings are held fixed. Table II reports the main configuration.

Evaluation and parsing details are reported in Section III-F. Main tables report three-seed means and standard deviations when available, and GSM8K additionally uses paired prediction-level bootstrap intervals for selected full-data comparisons.

### B. Protocol

The evaluation procedure is formalized as Protocol 1. It is a diagnostic protocol for evaluating subset-gain claims, not a new data-selection algorithm.

The protocol combines four checks: full-data recipe sanity, strong random or stratified-random baselines, difficulty-proxy diagnostics, and cost-accounting caveats. Its main output is not a selector leaderboard, but an evidence-qualified interpretation of whether a subset result is robust, recipe-sensitive, interval-limited, proxy-confounded, or cost-limited.

A key feature of the protocol is that it does not return only a score ranking. It also classifies the interpretation supported by each empirical observation. Table III defines these outcome categories. The categories are not mutually exclusive: for example, a subset result can be both recipe-sensitive and cost-limited. This makes the protocol more than a checklist, because it provides a rule for deciding whether an observation supports a robust selector gain, no subset-over-full claim, a recipe-sensitive result, an interval-limited match, a proxy-confounded result, or a cost-limited result.

TABLE III
PROTOCOL INTERPRETATION CATEGORIES.

| Category | Operational meaning |
|---|---|
| Robust subset gain | Subset remains above tuned full-data training and strong simple baselines. |
| No subset-over-full gain | Full-data training remains strongest under the tested conditions. |
| Recipe-sensitive gain | The ordering changes after full-data or subset-side recipe checks. |
| Interval-limited match | The subset is close to full, but the paired interval does not support superiority. |
| Proxy-confounded gain | The score is entangled with label, answer, or length artifacts. |
| Cost-limited gain | Performance is favorable, but token/compute matching or full overhead accounting is absent. |

**Protocol 1. Controlled evaluation of QLoRA subset-gain claims**

1) Train a fixed full-data baseline.
2) Train random and stratified-random subset baselines.
3) Train diagnostic scored or diversity-based subset conditions.
4) If a subset-over-full gain appears, run full-data recipe sanity checks.
5) Run subset-side sanity checks for the strongest subset condition.
6) Diagnose difficulty-proxy behavior using label distributions and length correlations.
7) Compare performance, interpretation changes, and whether cost or compute matching is sufficient for an efficiency claim.
8) Assign each observation to the categories in Table III.

**Output:** a subset-gain judgment, baseline comparison, proxy-confounding report, and cost caveats.

The protocol is intended as a minimum reporting standard rather than an exhaustive experimental prescription. Its checks can be applied selectively: recipe sanity checks are most important when a subset appears to match or exceed full-data training, proxy diagnostics are most important when a score such as normalized-NLL is used for selection, and cost caveats are required whenever performance is discussed together with efficiency. This design keeps the protocol practical while making the interpretation of subset-over-full observations explicit.

### C. Applying the Protocol in This Study

The protocol is applied in a staged manner. First, the fixed-recipe matrix is used to identify whether any apparent subset-over-full result appears. In this matrix, AG News produces the strongest apparent subset-over-full pattern, so it triggers the full-data and subset-side recipe sanity checks in Steps 4–5 of Protocol 1. BoolQ does not produce a subset-over-full claim under the tested conditions; it is therefore used primarily to test whether full-data training remains a strong baseline. GSM8K produces a different pattern: a moderate difficulty-diversity band is close to full-data training, so paired prediction-level comparisons are used to determine whether the difference is

distinguishable from full. Finally, normalized-NLL difficulty is audited across tasks to test whether the score behaves like a utility signal or instead tracks label, answer, or length artifacts.

This staged use of the protocol is important because the fixed-recipe matrix alone would encourage a selector ranking, whereas the protocol asks whether each apparent gain survives the checks required for interpretation.

### D. Selection and Diagnostic Conditions

The selection conditions are intentionally lightweight and reproducible. Random sampling provides a uniform subset baseline, while stratified random balances labels for AG News and BoolQ and uses diagnostic binning for GSM8K. Diversity selection serves as an embedding-coverage diagnostic, difficulty selection ranks examples by normalized NLL from a smaller scorer, and difficulty-diversity banding first restricts candidates to a difficulty percentile interval before applying diversity selection. The original band uses the 40–90 percentile range; ablations use 30–80, 40–90, and 50–95.

The scored diagnostic, denoted `d3_lite` in the run identifiers, is a lightweight two-stage heuristic rather than a reproduction of D3. Let $d_i$ denote the token-normalized NLL of example $i$, $\tau_{0.7}$ its pool-level 70th percentile, and $\sigma_d$ its standard deviation. Examples are ranked by $q_i = -|(d_i - \tau_{0.7})/(\sigma_d + 10^{-12})|$. We retain the top $\min(5k, N)$ candidates and select $k$ examples using seeded cosine-distance $k$-center greedy over all-MiniLM-L6-v2 embeddings. At budgets of 20% or above, the candidate multiplier covers the valid pool, so this condition effectively reduces to the diversity stage and is interpreted only as a diagnostic baseline.

### E. Recipe Sanity and Diagnostics

Recipe sanity checks are performed on AG News because the fixed-recipe matrix first produced the clearest apparent subset-over-full gain on that task. The protocol is a triggered diagnostic protocol rather than an exhaustive hyperparameter sweep: additional recipe checks are run when a fixed-recipe result would otherwise support a subset-over-full claim or a near-full claim requiring further interpretation. AG News therefore receives both full-data and subset-side recipe sanity checks. BoolQ is interpreted mainly as a no-subset-over-full case because full-data training remains strongest under the tested conditions. GSM8K is interpreted as a near-full, interval-limited case, so the additional diagnostic is a paired prediction-level comparison rather than an additional recipe grid. This design does not rule out different orderings under other recipes; it limits the claims to the tested diagnostic grid.

Difficulty diagnostics compute class or answer-label distributions across normalized-NLL difficulty deciles for AG News and BoolQ and length correlations for GSM8K. Official task scores are taken from the consolidated metric table. Prediction-level audits are used only as auxiliary diagnostics. Operational timing logs are retained only for audit bookkeeping; they are not used to rank methods by efficiency because selection overhead and compute matching are incomplete.

### F. Implementation and Statistical Details

Data processing and selection use seed 42; main comparisons use training seeds 1–3 on fixed subset identities. Means and standard deviations therefore describe training-seed variation, not uncertainty over all possible subsets. AG News and BoolQ use descriptive seed-level comparisons, whereas GSM8K additionally uses paired bootstrap intervals. Instruction-style prompts contain optional input and target output fields. Deterministic decoding uses sequence/new-token limits of 512/6 for AG News, 512/4 for BoolQ, and 768/128 for GSM8K; task-specific parsers normalize four topic labels, yes/no answers, and final numeric answers, respectively.

Random sampling uses a fixed seed; stratification preserves AG News and BoolQ labels and uses normalized-NLL bins for GSM8K. Diversity uses `all-MiniLM-L6-v2` embeddings with $k$-center greedy cosine coverage. Difficulty is token-normalized target NLL from Qwen2.5-0.5B-Instruct with prompt and padding tokens masked. Difficulty-diversity first filters to a specified NLL-percentile band and then applies the same diversity procedure. The separate two-stage scored diagnostic defined in Section III-D was stored under the run identifier `d3_lite`; neither it nor the percentile-band conditions is a reproduction of D3.

For GSM8K, selected conditions are compared against full-data training using paired bootstrap intervals over evaluation examples. Prediction records are first aligned by evaluation example identifier for each target condition and the corresponding full-data run. Let $c_i^{(t)}$ and $c_i^{(f)}$ denote binary exact-match correctness for the target condition and the full-data condition on example $i$. The paired point estimate is

$$\Delta = \frac{1}{N} \sum_{i=1}^{N} (c_i^{(t)} - c_i^{(f)}).$$

We resample the $N$ aligned evaluation examples with replacement and report percentile 95% intervals. The pooled analysis concatenates three seed-aligned prediction vectors ($N = 3957$) and uses 10,000 resamples with analysis seed 20260504. These intervals quantify example-level uncertainty on the fixed evaluation set, not uncertainty over all training runs, subsets, or recipes.

Experiments used an A100 40GB, but incomplete timing logs omit parts of startup, download, evaluation, and selection. We therefore make no token-matched, compute-matched, or fully overhead-accounted efficiency claim; favorable subset results remain cost-limited.

## IV. Results

### A. RQ1: Recipe Checks Change Subset-Gain Interpretation

Table IV summarizes the AG News interpretation shift. Under the fixed recipe, the full-data baseline reaches $0.8247 \pm 0.0138$ accuracy, while the 10% stratified-random subset reaches $0.8830 \pm 0.0012$. A full-data sanity check reaches $0.8878 \pm 0.0047$ at 2 epochs, showing that the fixed full baseline was under-tuned. The same subset reaches $0.8929 \pm 0.0014$ at 4 epochs, so it is also recipe-sensitive.

TABLE IV
AG NEWS RECIPE CHECKS (THREE SEEDS PER CONDITION).

| Condition | LR | Epochs | Acc. ± SD |
|---|---|---|---|
| Fixed full | $2 \times 10^{-4}$ | 3 | $0.8247 \pm 0.0138$ |
| Full sanity | $2 \times 10^{-4}$ | 1 | $0.8589 \pm 0.0021$ |
| Full sanity | $2 \times 10^{-4}$ | 2 | $0.8878 \pm 0.0047$ |
| Full sanity | $2 \times 10^{-4}$ | 4 | $0.8410 \pm 0.0059$ |
| Full sanity | $1 \times 10^{-4}$ | 3 | $0.8096 \pm 0.0029$ |
| Full sanity | $3 \times 10^{-4}$ | 3 | $0.8680 \pm 0.0026$ |
| Stratified 10% | $2 \times 10^{-4}$ | 1 | $0.8205 \pm 0.0013$ |
| Stratified 10% | $2 \times 10^{-4}$ | 2 | $0.8507 \pm 0.0043$ |
| Stratified 10% | $2 \times 10^{-4}$ | 3 | $0.8830 \pm 0.0012$ |
| Stratified 10% | $2 \times 10^{-4}$ | 4 | $0.8929 \pm 0.0014$ |

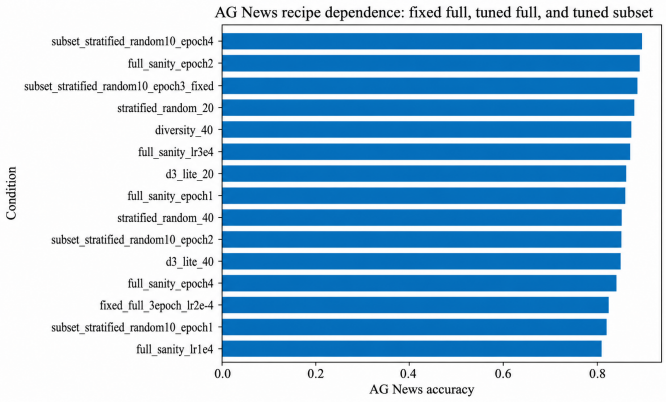



Fig. 1. AG News recipe dependence. Both the full-data baseline and the 10% stratified subset change under sanity checks.

Thus, the AG News result is best interpreted as a recipe-sensitive subset-over-full pattern within the tested grid, not as a robust selector gain. The recipe grid is not exhaustive, the subset and full-pool conditions are not token- or compute-matched, and AG News is summarized with seed-level variation rather than prediction-level paired intervals. The main lesson is therefore a recipe-sanity one: subset-over-full claims should not be made against a single fixed full-data recipe. The recipe-sanity results are reported as post-hoc diagnostic comparisons, not as a procedure for selecting a final deployed recipe from the evaluation split. Accordingly, we use this result to motivate the protocol, not to claim intrinsic superiority of stratified 10% selection.

### B. RQ2: Simple Baselines Are Mandatory Controls

The fixed-recipe comparison shows that simple baselines are not weak controls. AG News stratified random 10% exceeds the fixed full baseline and the best diversity condition; BoolQ favors full-data training; and GSM8K full-data has the highest mean while the 30–80 difficulty-diversity band is close. Random and stratified random are therefore mandatory controls.

Table V shows why simple baselines are central to the protocol. In the fixed-recipe matrix, the strongest AG News simple baseline is above the fixed full-data baseline and also above the strongest scored or diversity diagnostic condition. BoolQ shows the opposite pattern: full-data training remains strongest, and no subset condition should be interpreted as a full-data replacement. GSM8K lies between these cases: full-data training has the highest mean, but the moderate difficulty-diversity band is close enough to require paired analysis rather than a raw ranking. Thus, the fixed-recipe matrix does not support a single selector ordering; it motivates the subsequent interpretation checks.

TABLE V
FIXED-RECIPE COMPARISON OF FULL-DATA, SIMPLE, AND DIAGNOSTIC CONDITIONS. SCORES ARE MEANS OVER THREE TRAINING SEEDS.

| Task | Best full | Best simple | Best diagnostic |
|---|---|---|---|
| AG News | Full: 0.8247 | Strat. 10%: 0.8830 | Diversity 40%: 0.8703 |
| BoolQ | Full: 0.7587 | Random 40%: 0.7176 | Seeded $k$-center 40%: 0.7322 |
| GSM8K | Full: 0.3715 | Strat. 40%: 0.3563 | Diff.-div. 30–80: 0.3690 |

TABLE VI
DIFFICULTY-PROXY FAILURE MODES.

| Task | Low decile | High decile | Diagnostic |
|---|---|---|---|
| AG News | Sci/Tech 86.8% | Sports 89.8% | Class distribution is highly skewed. |
| BoolQ | True 98.4% | False 98.2% | Answer labels dominate decile extremes. |
| GSM8K | – | – | Length correlation: char 0.417, word 0.435. |

### C. RQ3: Normalized-NLL Difficulty Is Confounded

Difficulty selection appears to target informative examples, but Table VI shows why it must be audited. On AG News, the lowest difficulty decile is 86.8% Sci/Tech while the highest is 89.8% Sports. On BoolQ, the lowest decile is 98.4% true and the highest is 98.2% false. On GSM8K, difficulty correlates with character length ($r = 0.417$) and word length ($r = 0.435$).

These diagnostics indicate that normalized-NLL difficulty is confounded with class or answer-label distribution on classification-style tasks and with length on GSM8K, rather than serving as an unconfounded utility signal. This matters because difficulty-only selection performs poorly in the settings where the proxy is most confounded: high-difficulty regions are not necessarily high-utility regions.

The implication is that difficulty-based selection can change the composition of the training data for reasons unrelated to downstream utility. On AG News, difficulty deciles are strongly aligned with topic labels, so selecting by difficulty can implicitly select topics. On BoolQ, the decile extremes collapse toward opposite answer labels, so difficulty-based filtering can alter the yes/no prior. On GSM8K, the positive correlation with length suggests that difficulty can partially track verbosity or solution length rather than only mathematical usefulness. These effects do not imply that difficulty is useless, but they show that normalized-NLL difficulty must be audited before it is used as a selection score.

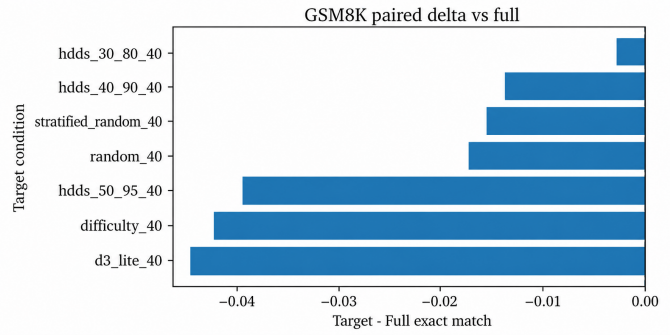



Fig. 2. GSM8K paired deltas against full-data training. Moderate difficulty-diversity banding is not distinguishable from full under the paired interval used here, while higher-difficulty bands are lower. Plot labels beginning with `hdds_` denote difficulty-diversity band diagnostics; `d3_lite_40` is the seeded diversity-stage diagnostic defined in Section III-D. These are run identifiers, not new selector proposals or D3 reproductions.

TABLE VII
GSM8K PAIRED COMPARISONS AGAINST FULL-DATA TRAINING.

| Condition vs. full | Delta | 95% paired interval |
|---|---|---|
| HDDS 30–80 band, 40% | -0.0028 | [-0.0164, 0.0104] |
| HDDS 40–90 band, 40% | -0.0136 | [-0.0268, -0.0008] |
| HDDS 50–95 band, 40% | -0.0394 | [-0.0531, -0.0258] |
| Difficulty 40% | -0.0422 | [-0.0556, -0.0283] |

### *D. RQ4: Cost Caveats and Paired Diagnostics Constrain the Claims*

Cost accounting constrains the practical interpretation of subset gains. Test performance alone can be insufficient for reliable comparison when development choices and computation budgets differ [20]. Because the AG News comparison is not token-matched, compute-matched, or fully overhead-accounted, its strongest subset result should be interpreted as a performance pattern rather than an end-to-end efficiency result.

GSM8K paired comparisons, plotted in Fig. 2, provide another diagnostic view. The HDDS 30–80 band at 40% differs from full-data training by -0.0028 exact match with a 95% bootstrap interval of [-0.0164, 0.0104], so it is not distinguishable from full under the paired bootstrap interval used here. Higher-difficulty bands and top-difficulty selection are lower, supporting the warning that useful difficulty is not equivalent to selecting the hardest examples.

Table VIII summarizes the main audit result. None of the main subset observations should be interpreted as an unconditional robust selector win. The strongest AG News result is recipe-sensitive and cost-limited; the BoolQ result supports no subset-over-full gain; the GSM8K result is interval-limited with cost caveats; and the difficulty-only conditions illustrate proxy confounding. The protocol therefore changes the output of the experiment from a selector ranking into an interpretation of what kind of evidence each observation supports.

## V. PRACTICAL LESSONS

The protocol yields four lessons for QLoRA data-selection studies.

TABLE VIII
EMPIRICAL OBSERVATIONS MAPPED TO PROTOCOL CATEGORIES.

| Observation | Protocol category | Interpretation |
|---|---|---|
| AG News 10% stratified subset beats fixed full | Recipe-sensitive; cost-limited | Highest mean within the tested grid, but not a robust subset-gain claim without recipe, statistical, and cost-accounting caveats. |
| BoolQ full-data training remains strongest | No subset-over-full gain | Full-data remains the appropriate reference. |
| GSM8K 30–80 difficulty-diversity band is close to full | Interval-limited; cost-limited | Competitive, but not a robust superiority claim. |
| High-difficulty selection degrades | Proxy-confounded | Difficulty can encode artifacts rather than utility. |

**Lesson 1: Do not compare subsets only against a fixed full-data recipe.** The AG News case shows that a fixed full-data baseline can be misleading, while the subset side can also be recipe-sensitive.

**Lesson 2: Treat random and stratified random as mandatory baselines.** Several complex or scored conditions do not clearly dominate simple baselines. Reporting only against a weak full-data baseline can overstate selector quality.

**Lesson 3: Audit difficulty before interpreting it as utility.** Normalized-NLL difficulty can reflect label distribution or length rather than example usefulness.

**Lesson 4: Separate performance from efficiency claims.** Subset performance should not be interpreted as an efficiency win unless token exposure, compute matching, selection overhead, and operational costs are reported consistently.

Taken together, these lessons imply a minimum reporting standard for QLoRA data-selection studies. A subset-over-full result should report: (i) whether the full-data baseline was checked under at least a minimal recipe sanity grid, (ii) whether random and stratified-random baselines were included at matched budgets, (iii) whether the scoring proxy was audited for label, answer, or length confounding, and (iv) whether performance, training cost, selection overhead, and token/compute matching were reported separately. Without these checks, a subset gain should be treated as an apparent gain rather than as evidence of selector superiority.

## VI. LIMITATIONS

This study targets evaluation reliability rather than state-of-the-art selector ranking. Accordingly, it uses lightweight, reproducible diagnostic conditions to expose failure modes under controlled settings. The main analyzed matrix focuses on Qwen2.5-1.5B with a Qwen2.5-0.5B pilot, and the tasks cover classification, reading comprehension, and mathematical reasoning but not all possible domains.

Because the goal is evaluation reliability rather than selector ranking, this study uses lightweight diagnostic conditions rather than full implementations of recent data-selection methods. Applying the protocol to selectors such as LESS, Data Whisperer, D3, T-SHIRT, and GIST is a natural next step [5], [12], [13], [24], [25]. The recipe sanity grid is also targeted rather than exhaustive, and the comparison is not token- or

compute-matched. For this reason, available training GPU-hour logs are retained only as coarse run logs rather than as compute-matched efficiency evidence, and all efficiency-related interpretations are made conservatively.

## VII. Conclusion

This paper presents a controlled evaluation protocol for QLoRA data selection. The empirical audit shows that apparent subset gains can be misattributed unless full-data recipes, simple baselines, difficulty proxies, and cost accounting are checked. AG News shows a recipe-sensitive subset-over-full pattern within the tested grid, BoolQ favors full-data training, and GSM8K shows a moderate difficulty-diversity band that is close to full but interval-limited. Difficulty diagnostics reveal label and length confounding. The main recommendation is practical: QLoRA data-selection studies should report not only which subset scores highest, but whether the claimed gain survives recipe, baseline, proxy, and cost-accounting caveats.

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